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Spacecraft Attitude Dynamics
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Project specifications:

	Assigned specification	Modifications (if any)	Motivation for modification
Platform	**Minisat**		
Attitude parameters	**Direction Cosines**		
Mandatory sensor	**Horizon Sensor**	**Add Magnetometer**	**Attitude determination**
Actuators	**Variable Thrust Jet**		

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1. Introduction

This report analyses the design and simulation of an Attitude Control System (ACS) for a minisatellite undertaking a nadir-pointing mission in Low Earth Orbit (LEO). The control system is actuated using variable jet thrusters, while attitude determination is achieved through a horizon sensor coupled with a magnetometer.

The mission is divided into three critical stages: detumbling, slew manoeuvre, and pointing. Each stage is simulated, and the robustness of the actuators and control logic is evaluated and discussed in detail. The Direction Cosine Matrix (DCM) and angular velocities are employed to model the system's behaviour accurately.

The control strategy follows a proportional approach throughout all manoeuvres. However, during detumbling, only the magnetometer is utilised, as the horizon sensor lacks the required responsiveness for this phase. Conversely, the slew and pointing manoeuvres are executed with full attitude determination, leveraging data from both sensors.

2. Mission Description & Requirements

The assignment was to control a minisatellite equipped with variable jet thrusters and an Earth horizon sensor. For these specifications, a nadir-pointing mission in a Low Earth Orbit (LEO) was chosen, with the specific orbital parameters of the mission, which were inspired by an existing spacecraft that is also performing a nadir pointing, and these parameters are:

Table 1: Orbital parameters of the mission.

Orbital Parameter	**Value**
Semi-major axis	6899.75 [Km]
Eccentricity	0.0005895 [-]
Inclination	5.0097 [°]
Right ascension of the ascending node RAAN	308.7725 [°]
Argument of perigee	280.9595 [°]
Initial true anomaly	0 [°]
Orbital Period	5703.6 [sec]

For complete attitude determination, relying solely on the Earth horizon sensor is insufficient, as it provides only one direction with respect to the satellite's frame, while in order to have a full attitude determination it is required to have the direction of two linearly independent vectors. Therefore, additional sensors were evaluated to meet the requirements of the mission. Star sensors, while highly

accurate, were discarded due to their high cost. Sun sensors were excluded because eclipses in orbit would limit its ability to determine the position of the sun, and eventually full attitude determination will be computed.

After considering the options, a three-axis magnetometer was selected. Magnetometers are cost-effective, provide good accuracy, and have a high acquisition frequency. Moreover, as the mission operates in a Low Earth Orbit with low inclination, the magnetic field is strong and not aligned with the Earth's direction, making it an effective choice for attitude determination.

2.1. Mission Phases

The current mission is divided into three phases: Detumbling, Slew Manoeuvre & Pointing/Tracking.

The detumbling phase searches to deliver the angular velocities of the spacecraft after insertion into the designated orbit to $\approx 0\, rad/s$ by using the designed actuators.

Once the detumbling phase is done and the target angular velocities are reached the Slew Manoeuvre phase will take place, whereby using the full knowledge of the attitude of the spacecraft the control logic will approximate the satellite to the target attitude.

The final Tracking phase aims to make the satellite follow and maintain the reference attitude.

2.2. Mission Requirements

Taking into consideration the specifications given in this project the objective of the satellite's mission is to perform coarse Earth pointing. Given that we are using variable jet thrusters, reach a high fidelity in the pointing of the satellite increases the complexity of the control system. While these thrusters allow for adjustable torque and attitude changes, they do not provide the same level of fine control that more precise actuators like reaction wheels or control moment gyroscopes offer. The control system design reflects these constraints, focusing on maintaining a stable attitude within a predefined angular tolerance (e.g., within 1-3 degrees) rather than achieving pinpoint accuracy. This is especially important for long-term operations where computational or mechanical complexities could increase if high-fidelity control is enforced.

2.3. Satellite's Geometry

The satellite to be used for the assignment corresponds to a Mini Sat (between 200-500 kg). Below is presented the principal inertia matrix of the satellite.

$$J = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \quad (1)$$

Where $I_{xx} = 4.208 * 10^2, I_{yy} = 4.922 * 10^3, I_{zz} = 4.399 * 10^3\ [kg * m^2]$.

3. Dynamics & Kinematics Modelling

The dynamics of the system are derived from the Euler Equations, which describe the attitude motion of a rigid body. By writing the fundamental equation we have:

$$\bar{h} = \begin{Bmatrix} I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z \\ I_{yx}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z \\ I_{zx}\omega_x + I_{zy}\omega_y + I_{zz}\omega_z \end{Bmatrix} \tag{2}$$

By expanding the previous expression and referring to the principal inertia axes we can simplify the system's dynamics as:

$$\begin{cases} I_x\dot{\omega_x} + (I_z - I_y)\omega_z\omega_y = M_x \\ I_y\dot{\omega_y} + (I_x - I_z)\omega_x\omega_z = M_y \\ I_z\dot{\omega_z} + (I_y - I_x)\omega_y\omega_x = M_z \end{cases} \tag{3}$$

Where the torques M_x, M_y, M_z represent the contributions due to the acting environment perturbations and due to the control torques given by the satellite's actuators.

To complete the rotational dynamics of the spacecraft we need to join the Euler equations with an attitude parametrization. These attitude parameters allow us to define the orientation of the spacecraft from one frame to the other, in our case from the Earth-centred inertial frame to the principal inertia frame (or body frame).

As given by the requirements of our mission, the attitude parametrization to be used is the Direction Cosines Matrix, where u, v & w represent the unit vectors of the body frame in the inertial frame and each row represents a new axis.

$$A_{B/N} = \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix} \tag{4}$$

Lastly, we desire to define how this attitude parameters change with time, and how they are related to the angular velocities described by the dynamics. From [1] we get that the kinematics for the Direction Cosines are given as:

$$\frac{dA}{dt} = \lim_{\Delta t \to 0} \frac{A(t + \Delta t) - A(t)}{\Delta t} = -[\bar{\omega}\Lambda]A(t) \tag{5}$$

Where the last term can be written as:

$$[\bar{\omega}\Lambda] = \begin{bmatrix} 0 & -\omega_w & \omega_v \\ \omega_w & 0 & -\omega_u \\ -\omega_v & \omega_u & 0 \end{bmatrix} \tag{6}$$

Finally, knowing the values of ω we can integrate the dA/dt to update the value of the matrix A.

Due to the accumulation of numerical errors, to guarantee the orthogonality of the A matrix we apply the following equation to orthogonalize at each integration step:

$$A_{k+1}(t) = A_k(t) * \frac{3}{2} - A_k(t) * A_k^{\,T}(t) * \frac{A_k(t)}{2} \tag{7}$$

4. Environment Modelling

This paragraph introduces the modelling of the environment system, which will be used to introduce the external perturbations to the control system.

4.1. Disturbance torques

For the evaluation of the environment disturbances to model in our satellite, it was decided to only consider the perturbances that had a major impact on the spacecraft. The perturbations were chosen by evaluating the torques based on the satellite's geometry described in 2.3. The values that were calculated are in Table 2:

Table 2: Maximum disturbance torques.

Disturbance Torques	**Gravity Gradient**	**Earth Magnetic Field**	**Solar Radiation Pressure**	**Aerodynamic Drag**
Maximum Calculated Values (N·m)	$8.2 * 10^{-3}$	$4.62 * 10^{-4}$	$3.22 * 10^{-6}$	$1.1 * 10^{-5}$

These values resulted in the elimination of solar radiation pressure as a significant factor, which was expected: the magnitude of the gravity gradient is justified by the satellite's dimensions, while the low Earth orbit (LEO) accounts for the dominance of the remaining forces.

The results for the worst-case scenarios were validated through simulations, which were conducted under conditions where the satellite operated without active control and with high initial angular velocities on all axes. This approach ensured that all possible attitude conditions were accounted for during the analysis.

4.1.1 Gravity Gradient

The gravity gradient perturbation was modelled using the equation described in Equation (8)**Error! Reference source not found.**:

$$\begin{cases} M_x = \dfrac{3GM_t}{r^3}(I_z - I_y)c_2c_3 \\ M_y = \dfrac{3GM_t}{r^3}(I_x - I_z)c_1c_3 \\ M_z = \dfrac{3GM_t}{r^3}(I_y - I_x)c_2c_1 \end{cases} \tag{8}$$

Where $\bar{c}$ is defined as follows:

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = A_{B/L} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{9}$$

In which the matrix $A_{B/L}$ was computed from the knowledge of matrix $A_{B/N}$ (Direction cosines matrix) and the Local Vertical Local Horizontal frame matrix $A_{L/N}$, which is given by Equation (10):

$$A_{L/N} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & \sin i \\ 0 & -\sin i & \cos i \end{bmatrix} \tag{10}$$

And then, $A_{B/L}$ is going to be computed as:

$$A_{B/L} = A_{B/N} * A_{L/N}^{T} \tag{11}$$

During the tracking the gravity gradient has a stabilizing effect due to the geometry and attitude of the satellite.

4.1.2 Aerodynamic drag torque

The velocity of the spacecraft relative to the atmosphere in the inertial frame was first determined. This process began by calculating the velocity vector in the perifocal frame, which was then rotated into the Earth-Centred Inertial (ECI) frame. Subsequently, the rotational velocity of the Earth was considered to determine the relative velocity. Finally, this velocity vector was transformed into the body frame by multiplying it with the Direction Cosine Matrix (DCM) of the spacecraft, yielding the velocity components in body coordinates.

The aerodynamic force of each face can be computed using Equation (12):

$$F_{a_i} = -0.5\, Cd\, \rho(h,t) v_r^2 \frac{v_r}{||v_r||} \left(n_s . \frac{v_r}{||v_r||} \right) A \tag{12}$$

Where *Cd=2.2*, assuming each face of the satellite behaves as a flat plate. The atmospheric density, ρ, was considered constant at 520 km due to the negligible fluctuations in altitude. Finally, the torque is computed as:

$$M = \sum r_i \wedge \mathrm{F_{ai}} \tag{13}$$

Where, the r_i is the vector between the centre of each face and the centre of gravity of the satellite.

4.1.3 Magnetic torque

The magnetic torque acting on the satellite was modelled by assuming a realistic magnetic dipole moment, *D*, for the satellite. The satellite was considered to be of class 2 as given by [1], and the dipole moment components defined as:

The magnetic torque, *M*, was calculated using the cross-product of the Earth's magnetic field, *B*, and the satellite's dipole moment, *D*, as follows:

$$M = B \times D \tag{14}$$

The computation of the magnetic field is the most challenging aspect of this modelling. Since the satellite operates in Low Earth Orbit (LEO), where Earth's magnetic field varies significantly, and given that one of the onboard sensors is a magnetometer, an accurate evaluation of the magnetic field is essential. To achieve this precision, a 13th-order magnetic field model was implemented based on the method described in [2]. This high-order model captures the complex variations of the Earth's magnetic field with high fidelity.

The implementation of the 13th-order magnetic field model enabled a direct comparison with a simplified dipole model as described in [1]. By analysing the resulting contour plots of the magnetic field, significant differences between the two models are apparent. These differences visually and quantitatively demonstrated the limitations of the simplified model and underscored the necessity of employing the complete 13th-order model. The enhanced accuracy of this model is crucial for reliable torque computations and precise satellite attitude control.

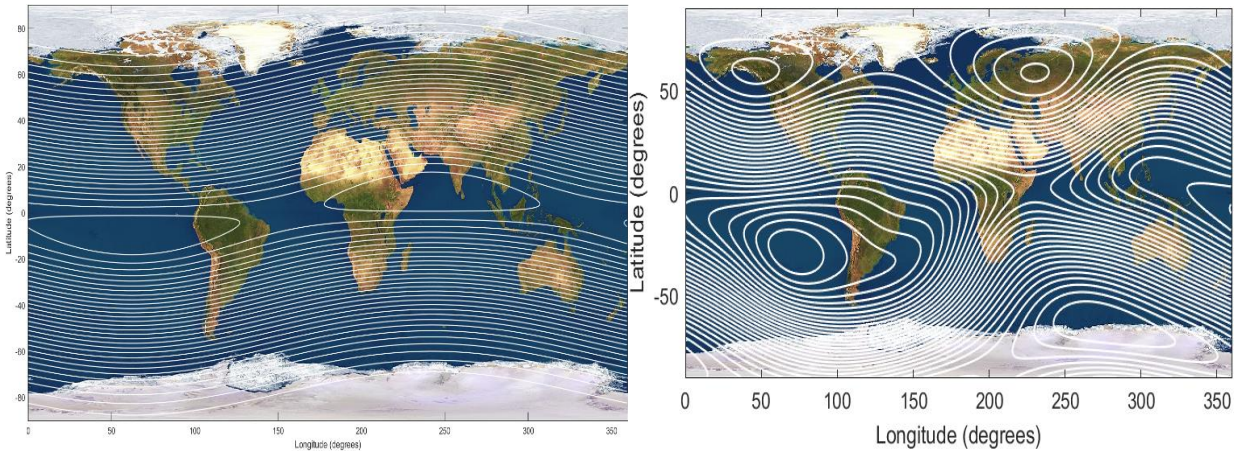


Figure 1: Dipole model v/s 13th order IGRF model.

4.2. Orbit propagation

The true anomaly was integrated using Equation (15), and it was used to compute the distance from the attractor in Equation (16). The forces and torques acting on the spacecraft were not considered during orbit propagation, as their effects are expected to be negligible over the duration of the simulation.

$$\dot{\theta} = \frac{n(1 + e\cos\theta)^2}{(1 - e^2)^{\frac{3}{2}}} \tag{15}$$

Finally, the position vector was calculated, accounting for the orientation of the orbital plane with respect to the Earth-Centred Earth-Fixed (ECEF) frame.

$$r = \frac{a(1 - e^2)}{1 + e\cos\theta} \tag{16}$$

$$\bar{r}_N = r \begin{bmatrix} \cos\theta \\ \sin\theta\cos i \\ \sin\theta\sin i \end{bmatrix} \tag{17}$$

Consequently, the orbital elements remain constant throughout the simulation.

5. Sensors

The outputs of the two devices were their specific directions in the body frame. To account for errors arising from non-idealities, a simple performance model was implemented. These included bias and random errors, tailored to the characteristics provided in the datasheets.

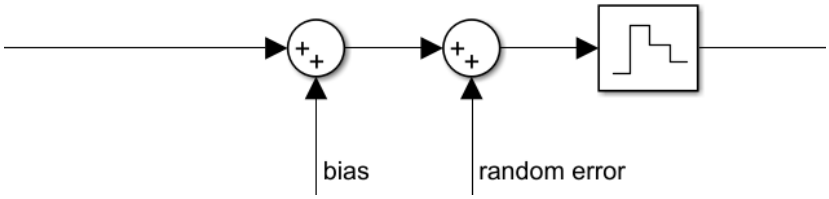


Figure 2: Simulink Bias Modelling.

The bias error was modelled as a constant, small rotation of the direction. For random errors, both Brownian noise (Brown noise) and white noise were incorporated. To ensure the error was random both in the magnitude of the angle between the real and measured vectors and in the direction of this rotation, the Rodrigues' formula was applied twice.

The first application of Rodrigues' formula rotated the initial vector around the z-axis, while the second rotation was performed around the axis of the real direction vector, ensuring complete randomness in the error.

The Rodrigues' rotation formula is expressed as:

$$v_{rotated} = v\cos(\theta) + (k \times v)\sin(\theta) + k(k \cdot v)(1 - \cos(\theta)) \quad (18)$$

Finally, the outputs were sampled at the specified acquisition rate of each device.

5.1. Horizon sensor

The **STD-16 Horizon Sensor** [3] developed by Sodern was selected due to its precision and suitability for Low Earth Orbit (LEO) missions. This sensor uses conical scanning technology to detect the Earth's horizon, providing accurate measurements of the satellite's pitch and roll. Its features make it ideal for applications requiring high accuracy and robust operation.

The sensor was chosen for its high precision, as its low bias error and noise ensure reliable measurements. Additionally, it is well-suited for Low Earth Orbit (LEO) missions, being optimized for altitudes between 300 km and 6,000 km.

The main specifications of the STD-16 sensor are summarized in Table 3:

Table 3: Horizon Sensor Specifications.

Parameter	**Value**
Altitude	300 km - 6,000 km
Bias Error	0.06° (3σ)

Typical Noise	0.042° (3σ)
Data Rate	1 Hz
Supply Voltage	20 - 52 V
Dimensions	175 mm x 208 mm x 386 mm
Mass	3.5 kg

As a simplification, the horizon sensor is assumed to have a complete field of view (FOV). This functionality is required only during the slew manoeuvre, as it is the only phase where the initial attitude is not known. During the tracking phase, the Earth remains consistently within the sensor's actual field of view.

5.2. Magnetic sensor

The **Hi-Rel Magnetometer FGM-A-75** [4] was selected due to its high reliability and precision in measuring the Earth's magnetic field, which is crucial for attitude determination in Low Earth Orbit (LEO) missions. This high-performance magnetometer is designed to deliver accurate data for orientation control in space, providing high sensitivity and low noise measurements.

The sensor was chosen for its robust precision, low bias errors, and high reliability in space environments. Its ability to operate effectively in LEO ensures it meets the mission's requirements, providing accurate data across a range of altitudes.

The main specifications of the FGM-A-75 sensor are summarized in Table 4:

Table 4: Magnetometer Specifications.

Parameter	**Value**
Angular Accuracy	0.1° (typical)
Data Rate	Up to 100 Hz
Magnetic Field Range	±75,000 nT (±75 µT)
Noise Level	0.1 nT (typical)
Weight	0.4 kg
Tolerance	Low bias error (±0.5 nT)

6. Attitude determination

The algebraic method was chosen to carry out the attitude determination, to find through the inputs of the 2 chosen sensors the real matrix $A_{B/N}$.
By using the independent unitarian vectors measurements of the Horizon Sensor in inertial frame $\widehat{\widehat{r_N}}$ and in body frame $\widehat{\widehat{r_b}}$ and the measurements of the magnetic field obtained by the magnetometers in

inertial frame $\widehat{B_N}$ and in body frame $\widehat{B_b}$ we compute the following vectors with Equations (19), (20) & (21):

$$s_1 = \hat{r}_b \qquad v_1 = \widehat{r_N} \tag{19}$$

$$s_2 = \frac{\hat{r}_b \times \widehat{B_b}}{\left\|\hat{r}_b \times \widehat{B_b}\right\|} \qquad v_2 = \frac{\widehat{r_N} \times \widehat{B_N}}{\left\|\widehat{r_N} \times \widehat{B_N}\right\|} \tag{20}$$

$$s_3 = \hat{r}_b \times s_2 \qquad v_3 = \widehat{r_N} \times v_2 \tag{21}$$

The unitarian vectors s_1, s_2, s_3 are orthogonal to each other and assembled into the matrix S=[s_1 s_2 s_3] and the same applies to the unitarian vectors v_1, v_2, v_3 by building the matrix V=[v_1 v_2 v_3].

This way the Direction Cosines Matrix is given by the following equation.

$$A_{B/N} = SV^T \tag{22}$$

7. Actuators

7.1. Actuator configuration

The actuators given are the variable jet thrusters. To have a complete control a simple configuration was chosen as presented in Figure 3 with 4 thrusters.

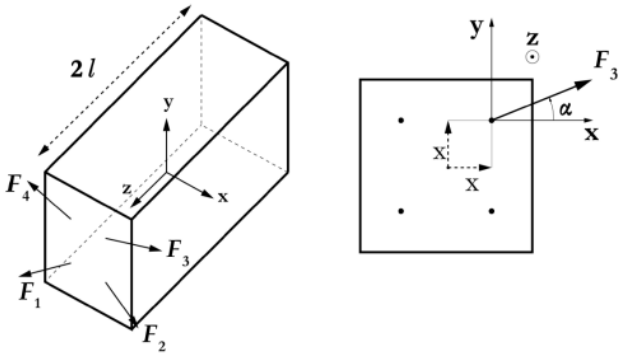


Figure 3: Thrusters Configuration.

Where the physical characteristics are $\alpha = \frac{\pi}{6}$; $l = 1.2$.

The torque will be given Equation (23):

$$\overline{\boldsymbol{T}}_{\boldsymbol{tot}} = \left[\widehat{\boldsymbol{R}}\right]\overline{\boldsymbol{F}} \tag{23}$$

$$\left[\hat{R}\right] = \begin{bmatrix} l\sin\alpha & 0 & 0 \\ 0 & l\cos\alpha & 0 \\ 0 & 0 & x\sin\alpha - x\cos\alpha \end{bmatrix}\begin{bmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \tag{24}$$

To compute the forces required to achieve a requested torque, the pseudoinverse method can be used. However, since thrusters can only generate positive forces, the null space vector is employed to ensure that the computed forces are both realistic and non-negative.

$$\bar{F} = [\hat{R}] * u_{ideal} + \gamma w \tag{25}$$

Where γ is a scalar defined by Equation (26):

$$\gamma = max_{i=1,\dots,N}([\hat{R}] * u_{ideal})_i / w_i \tag{26}$$

In this configuration, $w > 0$ is the null space vector. Therefore, it is sufficient to add this vector, scaled by the absolute value of the minimum force, to the force vector computed using the pseudoinverse.

7.2. Gas thrusters

The variable gas thruster selected for this application is the "MR-103G 1N" [5] due to its ability to provide low magnitudes of thrust that can reach a minimum of 0.19N in a steady state thrust, and at a high frequency. This capability makes it ideal for our requirements during all stages of our mission, detumbling, slew manoeuvre, and pointing. An additional advantage of this thruster is its low fuel consumption. Where it needs a maximum fuel rate of 0.5g/s, which not only reduces the overall mass of the spacecraft but also extends its operational lifespan.

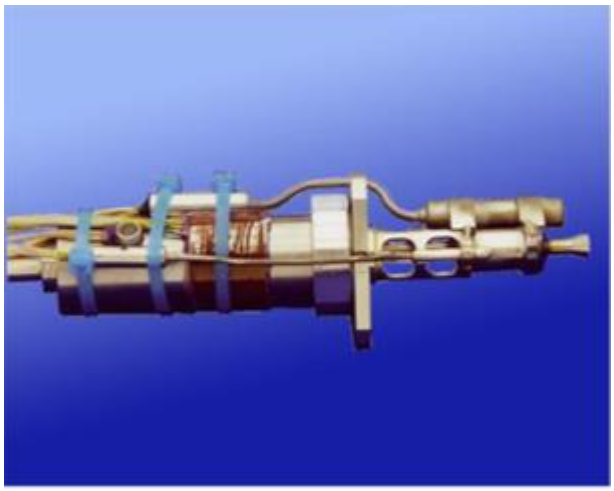


Figure 4: MR-103G 1N Variable Gas Thruster

The main technical specifications of the "MR-103G 1N" thruster is in Table 5:

Table 5: MR-103G 1N Thruster Specifications.

Parameter	Value
Propellant	Hydrazine
Thrust steady state	0.19-1.13 [N]
Flow rate	0.09-0.5 [g/s]
Mass	0.33 [Kg]

7.3. Gas thrusters Modelling

For the gas thrusters modelling, the force and torque were calculated by using Equations (23) to Equation (26). In addition, a saturation block was added, having a main role of setting the minimum and maximum values that the actuators are able to deliver for a steady state thrust, and these limits depend on the actual constraints of the thruster mentioned in Table 5, so the minimum is "0.19" and maximum "1.13".

To approach more to the real behaviour of the thrusters, a random error was also added to the block of the actuators, having a value of 5% of the minimum thrust delivered by the thruster. In addition to a rate limiter, whose main role is to limit the random change of thrust value, having as limits -0.5 and 0.5. Note that these values were randomly chosen after a series of trial and error.

8. Control Logic

As stated in the introduction the mission is divided in three stages: Detumbling, Slew and Tracking, each one requires the implementation of a distinct control logic that, known the necessary attitude parameters, computes the required torque for control. The system to be controlled is by nature nonlinear, but thanks to some reshaping of the requirements that will be explained in the following paragraphs, all the controls are linear, and take the form given by Equation (27):

$$\bar{u} = K * \bar{x} \tag{27}$$

The state denoted with $\bar{x}$ is the set of angular velocities in the detumbling and the full attitude is added during the slew and tracking. The matrix K is a diagonal matrix that is either a 3x3 or 3x6, so that it always outputs a 3x1 output vector, the $\bar{u}$**.** This vector represents the ideal torque requested, this value that is the input of the Actuator logic. The control is then only proportional in the case of the detumbling, instead it is proportional and derivative in the slew and tracking phases, with a set of k_p and k_d coefficients, tuned for each task.

In the following three paragraphs the distinct control logics and the reasoning behind them are explained in detail.

8.1. Detumbling

The detumbling is the first manoeuvre, since it brings the satellite to a known and close to zero set of angular velocities, starting from an unknow, and possibly not small, set of starting rotations. Launcher, usually, guarantee certain ranges of accuracy during deployment, both in the geometrical pointing of the satellite and it's starting angular velocities. Nonetheless, this does not mean that a detumbling isn't needed, because the ranges vary in function of which axis is considered, and are also dependent on launcher integration, that is outside the scope of the project.

During the detumbling phase the only available sensor of the system is the magnetometer because it has a complete spherical FOV, without any preliminary assumption needed. This sensor gives as measurement the local magnetic field vector $\overline{B_b}$ in the body frame, the variation of which can be used to estimate the full set of angular velocities, thanks to Equation (28):

$$\dot{\overline{B_b}} = \overline{B_b} \times \omega \tag{28}$$

This calculated set of velocities is an estimation that is based on the premise that the change in local magnetic field due to the movement of the satellite around its orbit is negligible. The full formula would be as in Equation (29):

$$\dot{\overline{B_i}} = \dot{\overline{B_b}} + \omega \times \overline{B_b} \tag{29}$$

To consider this time dependent change in magnetic field as small enough to be noise, the manoeuvre needs to be quick.

For the proposed detumbling control, a variation of the B-dot method is proposed, by rearranging the Equation (30).

$$\dot{\overline{B_b}} = \overline{B_b} \times \omega = \overline{B_b} \times \omega_{\perp} \tag{30}$$

Where $\omega_{\perp}$ corresponds to the projection of ω in the direction orthogonal to $\overline{B_b}$. Then, to obtain the expression for $\omega_{\perp}$ we can write the following expression.

$$\omega_{\perp} = \frac{\left(\dot{\overline{B_b}} \times \overline{B_b}\right)}{\|\overline{B_b}\|^2} \tag{31}$$

Given that we are working with variable jet thrusters, by adopting the B-dot method we obtain that the control law would be:

$$u_c = -K_{gain} * \omega_{\perp} \tag{32}$$

Where $K_{gain} \in \mathbb{R}^+$ corresponds to the control gain and is a constant parameter to be defined in accord to the specifics of the mission.

Since we are using variable jet thrusters as actuators, the convergence of the detumbling phase to angular values close to 0 for the three axes must be performed in a short amount of time due to fuel constrains.

8.2. Slew manoeuvre

The slew manoeuvre is executed following the detumbling phase, where the satellite's initial conditions are characterized by minimal angular velocities and a random initial attitude. During this phase, both onboard sensors, including the Horizon sensor, are operational since the latter is no longer limited by latency issues. The control strategy is consequently based on the satellite's full attitude, with angular velocities estimated using finite differences of the measured attitude angles.

The objective of the manoeuvre is to achieve a predefined final attitude within a constrained time frame of 100 seconds. Since the satellite's mission involves inertial nadir pointing, the satellite's body axes must align with the Local Vertical Local Horizontal (LVLH) frame at the end of the manoeuvre. An algorithm determines the target LVLH frame corresponding to 100 seconds after the manoeuvre’s initiation.

Knowing both the initial and final attitudes, the manoeuvre is implemented using a control law proportional to both the angular velocities and the angular differences between the desired and current attitudes. However, directly applying this control at the start could lead to abrupt changes in angular velocity due to significant initial attitude differences. Such transitions may result in mechanical stress, excessive fuel consumption, or damage to satellite components.

8.2.1 Control logic

To mitigate these risks, the manoeuvre generates a sequence of intermediate attitudes that smoothly transition from the initial to the final state. The process begins with the computation of the Direction Cosine Matrix (DCM) of the relative rotation, R, between the initial frame A and the final frame B, as described in [6]:

$$R = AB^T \tag{33}$$

Once R is determined, the Euler axis and Euler angle of the rotation are calculated. These parameters are then used to generate a set of intermediate frames, where the Euler axis remains constant, and the rotation angle varies progressively from 0 to the total Euler angle.

The evolution of the rotation angle over time is obtained sending the step like change in angle to a first-order transfer function. This function is specifically tuned to ensure that 99.91% of the total rotation angle is achieved within 500 seconds, facilitating a smooth transition.

The slew manoeuvre concludes after 500 seconds, at which point the satellite switches to its tracking mode. The results are assessed to verify proper alignment with the Local Vertical Local Horizontal (LVLH) frame and the successful completion of the manoeuvre.

8.3. Tracking

The tracking control follows, as the name implies, a reference attitude, that changes in time. In the case of the actual tracking phase, the goal is nadir pointing, so the reference attitude is the LVLH (local vertical local horizon), that can be easily computed as a DCM in function of the position in orbit.

The satellite is expected to maintain an attitude close enough to the requested one to allow the implementation of a simple linear, proportional derivative, control logic. The same reasoning is applied to the slew manoeuvre, to justify the usage of the same K matrix, since the smooth change of the requested attitude during the manoeuvre is like the one of the LVLH.

In both phases the full attitude is known thanks to both sensors being in use, and the angular velocities are obtained through the finite difference. The attitude error is computed at each step with Equation (34):

$$A_{LVLH} \cdot A = \begin{bmatrix} 1 & \alpha_z & -\alpha_y \\ -\alpha_z & 1 & \alpha_x \\ \alpha_y & -\alpha_x & 1 \end{bmatrix} \tag{34}$$

In the case of the slew manoeuvre the A_{LVLH} is instead the current requested intermediate attitude.

9. Results of statistical analysis

The strategy used to validate the implemented control logics, for all parts of the mission, is a statistical analysis, that evaluates enough random initial conditions, and simulates the full mission, saving relevant data.
The initial conditions that must be set are the initial angular rates, the ranges for both taken from [7] and are:

Table 6: Initial conditions of Detumbling

	X	Y	Z
Angular rates	0.01 [rad/s]	0.2 [rad/s]	0.01 [rad/s]

Then for each phase of the mission the selected relevant data is the following:

9.1. Detumbling

For this manoeuvre, the most critical parameters to validate its efficacy are the norm of the angular velocity at the end of the manoeuvre and the time required for convergence. A statistical analysis was conducted on these two values across 100 initial conditions.

The initial conditions featured random angular velocities, with their maximum values varying by axis. Specifically, the deploy axis had a maximum angular velocity of 0.2 rad/s, while the other axes were limited to a maximum of 0.001 rad/s. These values were selected by referring to "Vega-C" handbook.

Similarly, the initial attitude was defined as a random rotation about the deploy axis, with a maximum angular deviation of 10 degrees, as specified in the same launcher manual.

The results presented in the table below show its effectiveness:

Table 7: Results of the statistical study of the detumbling.

mean residual angular velocity	0.2105 [deg/s]
maximum residual angular velocity	0.8048 [deg/s]
mean time of detumbling	148.39 [s]
maximum time of detumbling	711.1 [s]

9.2. Slew Manoeuvre

A similar statistical analysis was conducted on the relative rotation of the body axes at the end of the manoeuvre with respect to the desired attitude. The initial conditions for this analysis were taken as the output of the detumbling manoeuvre. To evaluate the effectiveness of the manoeuvre, the parameter chosen was the Euler angle of the relative rotation between the final and desired attitudes.

Table 8:Results of the statistical study for the slew

Mean Euler angle	1.3366 [deg]
Maximum Euler angle	2.4148 [deg]

9.3. Tracking

The tracking manoeuvre effectiveness is tested. In this case, the objective was to check that the deviation of the pointing with respect to the desired attitude remained within with a confidence level. A statistical analysis was conducted on the deviation angle over three orbits to verify these criteria.

Table 9: Results of the statistical results for the controlled tracking.

Mean x-Euler angle	0.3301 [deg]
Maximum x-Euler angle	1.0023 [deg]
Mean y-Euler angle	0.6732 [deg]
Maximum y-Euler angle	1.2179 [deg]
Mean z-Euler angle	0.6142 [deg]
Maximum z-Euler angle	1.4353 [deg]

To make sure that the spacecraft is not already in a stable condition, a small test was conducted where the values of the Euler angles for the tracking were checked, without the input of the control logic, and these were the values:

Table 10:Results of the statistical study for the uncontrolled tracking.

Mean x-Euler angle (uncontrolled)	14.6445 [deg]
Maximum x-Euler angle (uncontrolled)	36.3594 [deg]
Mean y-Euler angle (uncontrolled)	13.2598 [deg]
Maximum y-Euler angle (uncontrolled)	30.5160 [deg]
Mean z-Euler angle (uncontrolled)	27.5026 [deg]
Maximum z-Euler angle (uncontrolled)	36.7174 [deg]

As the values in the table show, the spacecraft is not already in a stable condition, and this verifies the need and the efficacity of the implemented control logic.

10. Conclusions

The statistical analysis conducted on the detumbling, slew, and tracking manoeuvres demonstrates the effectiveness of the implemented control system for attitude control.

For the detumbling manoeuvre, the results show that the control system efficiently reduces the angular velocity, with a mean residual angular velocity of 0.2105°/s and a maximum residual angular velocity of 0.8048°/s. The system also achieves reasonable convergence times, with a mean detumbling time of less than half an orbit. These values suggest that the system successfully stabilizes the spacecraft in a relatively short period, which is the purpose of using variable jet thrusters as they can deliver high amounts of torque in a short period of time.

For the slew manoeuvre, the analysis reveals the spacecraft can achieve the rotation from its initial detumbled attitude to the desired target attitude with minor deviations, confirming the manoeuvre's efficiency in achieving the target orientation.

The tracking manoeuvre results show that the control system maintains the spacecraft's attitude with a high degree of accuracy for a coarse earth pointing mission. The mean Euler angles across the three axes are relatively small: 0.3301° (x-axis), 0.6732° (y-axis), and 0.6142° (z-axis), with the maximum deviations also remaining within acceptable limits (1.0023° for x, 1.2179° for y, and 1.4353° for z). These results demonstrate that the system ensures the spacecraft remains within a tight pointing tolerance, even during extended operations.

Additionally, the analysis of uncontrolled tracking shows significantly higher deviations in Euler angles, with maximum values reaching up to 36.7° along the z-axis. This further reinforces the conclusion that the designed control system allows the spacecraft to adopt a stable configuration for long nadir-pointing missions.

11. References

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